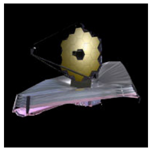

4598 - Silver Bullet for Dark Matter

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRCam Prime				
	1	NIRCam	NIRCam Imaging	(1) BulletCluster
NIRSpec Prime				

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
	2	AN_TOPLRD_50x50_0.04_NoMC_manualedit_demoted_nostuckopen	NIRSpec MultiObject Spectroscopy	(10) TOPLRD

ABSTRACT

The principal objective of this proposal is to study the nature of dark matter by using a merging galaxy cluster. Gravitational lensing provides a unique tool transforming these clusters into dark matter laboratories. The pioneering example, the Bullet Cluster, is an ideal laboratory for detecting dark matter and distinguishing between cold dark matter (CDM) and other scenarios (e.g., self-interacting dark matter). With this proposal, the limits on the dark matter self-interaction cross-section will be improved (by a factor of 4, to <0.2cm^2/g) to the extent that the possibility that dark matter self-interactions might explain the observed non-cuspy mass profiles can be fully excluded. To pin down the position of the dark matter component we require high resolution, absolutely calibrated mass maps that will be compared to simulations. The combination of weak and strong lensing measurements is needed to attain this goal. This can only be achieved with the excellent resolving power and spectroscopic capabilities of JWST. We therefore request NIRCam and NIRSpec observations of the Bullet Cluster. The combination of constraints from multiply lensed images (identified via color information and spectroscopy) and high-resolution weak lensing data will allow us to construct, self-consistently, the mass distribution from the very center to the outskirts. With this proposal we will further improve the selection of cluster members and measure their velocity to unambiguosly separate the baryonic and DM components. The proposal will enable investigations of the nature and properties of dark matter, and how it shapes galaxies and galaxy clusters and their evolution through cosmic time.

OBSERVING DESCRIPTION

The aim of these observations is to perform NIRCam imaging and NIRSpec spectroscopy of the Bullet Cluster.

1. NIRCam Imaging with NIRISS on a parallel field.

One NIRCam module, using a FULL dither pattern, to cover the detector gaps due to the geometry and simultaneous imaging of the subcluster. Filters as follows SW: F090W, F115W, F150W, F200W and LW: F277W, F356W, F410M, F444W. The NIRISS WFSS parallels use the F115W filter with both grisms and the F150W and F200W filters with only one grism. We understand observing in parallel with one grism in these 2 filters will lead to a higher fraction of targets affected by spectral contamination, however based on experience with NIRISS data such approach is maneagble.

2. NIRSpec Multi-Object R=100 Spectroscopy on the cluster field with NIRCam imaging on a parallel field

The NIRSpec multi-object spectroscopy will be targeted with a combination NIRCam pre-imaging obtained in this program (1. above) and of known sources from existing HST observations. High-priority targets are galaxies that are multiply imaged based on NIRCam imaging. The catalogs and MSA configurations in this APT file are therefore just placeholders to be changed when the NIRCam pre-imaging is obtained and processed. For NIRSpec Prime we will use NIRCAM parallels with SW: F090W, F150W, F200W and LW: F356W, F410M, F444W.

3. ORIENT, background and schedule constraints:

We require at least 60 days between the NIRCam pre-imaging and NIRSpec observations to allow for data processing, analysis and target selection.

We have analysed whether the observations are background-limited and should require the setting of a "Background Limited" special requirement following the procedure outlined on JDOX. All of the NIRCam observations are background limited and we set this special requirement with a limit of "Background no more than 20% above minimum". There is no need for such a requirement with NIRSpec.

There is an ORIENT (V3PA) constraints on the NIRCAM 10 20 deg and NIRSpec observations 85 105 for NIRSpec. Observations are schedulable given all constraints.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	BulletCluster	RA: 06 58 27.4418 (104.6143408d) Dec: -55 56 53.46 (-55.94818d) Equinox: J2000		
Comments: This object was generated by the targetselector and retrieved from the NED database. Category=Clusters of Galaxies Description=[Rich clusters]				
(10)	TOPLRD	RA: 06 58 28.0219 (104.6167579d) Dec: -55 56 50.91 (-55.94747d) Equinox: J2000		
Comments: Description=[]				

Proposal 4598 - Observation 1 - Silver Bullet for Dark Matter

Observation

Proposal 4598, Observation 1: NIRCam

Tue Feb 11 16:00:24 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Coordinated Parallel Template(s): NIRISS Wide Field Slitless Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	BulletCluster	RA: 06 58 27.4418 (104.6143408d) Dec: -55 56 53.46 (-55.94818d) Equinox: J2000		

Comments: This object was generated by the targetselector and retrieved from the NED database.
Category=Clusters of Galaxies
Description=[Rich clusters]

Template

NIRCam Imaging	NIRISS Wide Field Slitless Spectroscopy
Module: ALL	
Subarray: FULL	
Target Placement: Module Gap	

Dithers

#	Primary Dither Type	Primary Dithers	Dither Size	Subpixel Positions	Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
1	FULLBOX	6		1	NIRCam Only	NO_DITHERING

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W	F277W	SHALLOW4	7	1	1	1	365.05	
2	F150W	F277W	DEEP8	5	1	6	6	5669.015	
3	F150W	F277W	SHALLOW4	7	1	1	1	365.05	
4	F200W	F356W	SHALLOW4	7	1	1	1	365.05	
5	F200W	F356W	DEEP8	5	1	6	6	5669.015	
6	F200W	F356W	SHALLOW4	7	1	1	1	365.05	
7	F115W	F444W	SHALLOW4	7	1	1	1	365.05	
8	F115W	F444W	DEEP8	5	1	6	6	5669.015	
9	F115W	F444W	SHALLOW4	7	1	1	1	365.05	
10	F090W	F410M	SHALLOW4	7	1	1	1	365.05	
11	F090W	F410M	DEEP8	5	1	6	6	5669.015	
12	F090W	F410M	SHALLOW4	7	1	1	1	365.05	

Proposal 4598 - Observation 1 - Silver Bullet for Dark Matter

Spectral Elements

NIRISS Wide Field Slitless Spectroscopy	Exposure Type	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Two Extra Dithers	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	DIRECT	F115W		NIS	8	1	NO	1	1	354.313	
2	GRISM	F115W	GR150R	NIS	21	1	6	6	5475.753		3
DIRECT	F115W		NIS	8	1	NO	1	1	354.313		4
DIRECT	F115W		NIS	8	1	NO	1	1	354.313		5
GRISM	F115W	GR150C	NIS	21	1	6	6	5475.753		6	DIRECT
F115W		NIS	8	1	NO	1	1	354.313		7	DIRECT
F150W		NIS	8	1	NO	1	1	354.313		8	GRISM
F150W	GR150C	NIS	21	1	6	6	5475.753		9	DIRECT	F150W
	NIS	8	1	NO	1	1	354.313		10	DIRECT	F200W
	NIS	8	1	NO	1	1	354.313		11	GRISM	F200W
GR150C	NIS	21	1	6	6	5475.753		12	DIRECT	F200W	

Special Requirements

Aperture PA Range 9.92542306 to 19.92542306 Degrees (V3 10.0 to 20.0)
No Parallel Attachments
Background Limited. Background no more than 20th percentile above minimum

2 After 1 by 60 Days to <None specified>

Observation

Proposal 4598, Observation 2: AN_TOPLRD_50x50_0.04_NoMC_manualedit_demoted_nostuckopen Tue Feb 11 16:00:24 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Coordinated Parallel Template(s): NIRCam Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	TOPLRD	RA: 06 58 28.0219 (104.6167579d) Dec: -55 56 50.91 (-55.94747d) Equinox: J2000		

Comments:
Description=[]

Acquisition

NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	Filter: CLEAR; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

NIRSpec MultiObject Spectroscopy

TA Method: MSATA

HFF Readout Mode: false

Obtain Confirmation Images: No

Science Aperture: MSA Center

Primary Candidate List: TOPLRDPRimaries (86 sources)

Filler Candidate List: TOPLRD (1727 sources)

Spectral Overlap Map: jwst-nirspec-prism

Spectral Overlap Threshold: 1.5

NIRCam Imaging

Module: ALL

Subarray: FULL

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	7101725	104.617191	-55.962700	23.27925702078227	1	7104731	104.618920	-55.944916	24.535205266803093
1	7103316	104.542745	-55.953007	23.668273253125363	1	7105977	104.557005	-55.937674	23.14692822041743
1	7103337	104.549144	-55.952793	23.73191672076213	1	7106639	104.557516	-55.933114	24.615324722325223
1	7103362	104.648428	-55.952622	24.18592369522117	1	7109939	104.555106	-55.946719	23.184755902885072

Dithers

#	Dither Type
1	NONE

Spectral Elements

NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	104.59831675000001 Degrees - 55.94512805555553 Degrees	223.58987228228287			3	3	3545.1
2	1 (PRISM/CLEAR)	c2	3 Shutter Slitlet	104.6061975 Degrees - 55.95633555555554 Degrees	223.5833667738723			3	3	3545.1
3	1 (PRISM/CLEAR)	c3	3 Shutter Slitlet	104.60530558333335 Degrees - 55.94638250000003 Degrees	223.5840851201018			3	3	3545.1

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F090W	F410M	DEEP8	5	1	3	3	2834.507	
2	F200W	F356W	DEEP8	5	1	3	3	2834.507	
3	F115W	F444W	DEEP8	5	1	3	3	2834.507	

Special Requirements

Aperture PA Range 223.5745697 to 243.5745697 Degrees (V3 85.0 to 105.0)
No Parallel Attachments
MSA Scheduled Aperture PA 223.5746 to 223.5746 Degrees (V3 85.0 to 85.0)

2 After 1 by 60 Days to <None specified>